



VENTURE GLOBAL CALCASIEU PASS, LLC CONTROL VALVES DATASHEETS UNIT 139 – POTABLE WATER SYSTEMS

Oil, Gas & Chemical

1	14-SEP-2020	Issued for Purchase	MAM	MNA	EAC
0	30-JUL-2020	Issued for Purchase	MAM	MNA	EAC
B	12-JUN-2020	Issued for Approval	MAM	MNA	EAC
A	11-JUN-2020	Issued for Review	MAM	MNA	EAC
REV	DATE	DESCRIPTION	ORIG	CHK	APPR

PROJECT	DOCUMENT NUMBER	REVISION
Venture Global Calcasieu Pass, LLC	CP-139000-ICS-DAS-KIE-00001	1

Requisition number: CP-100000-ICS-REQ-KIE-0 P. O. number:

Rev. No.	Item Number	Tag Number	Instrument Type
1	1	139-000-PCV -39051	Pressure Valve Self-Actuating
1	2	139-000-PCV -39052	Pressure Valve Self-Actuating
1	3	139-000-PCV -39054	Pressure Valve Self-Actuating
1	4	139-000-PCV -39055	Pressure Valve Self-Actuating
1	5	139-000-PCV -39056	Pressure Valve Self-Actuating
1	6	139-000-PCV -39057	Pressure Valve Self-Actuating
1	7	139-000-PCV -39058	Pressure Valve Self-Actuating
1	8	139-000-PCV -39059	Pressure Valve Self-Actuating

Domain: VG_CALCASIEU_PASS				Plant: Calcasieu Pass LNG			
Area: 1				Unit: 39 - Potable Water Systems			
				Binder Package		VENTURE GLOBAL LNG  Kiewit	
1	MM	14-Sep-2020	Issued for Purchase	Formal issue Date: 14-Sep-2020			
0	MM	30-Jul-2020	Issued for Purchase	Instrument Specification List Control Valves, Unit 139			
B	MM	12-Jun-2020	Issued for Approval				
A	MM	11-Jun-2020	Issued for Review			Page 1 of 1	
No.	By	Date	Description	Doc. No.: CP-139000-ICS-DAS-KIE-00001			Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39051				
	2	Service			POT WAT TO WAT TRTMT BLDG				
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001				
	4	Location	Area Class		INLINE		Class I, Div 2, Group C & D, T3		
	5	Project Name	Project Number		Calcasieu Pass LNG		103930		
	6	Control Number	Work Breakdown Structure						
	7	Line Number			2"-139-000-PTW39030-1K0K3-1.0"HET				
	8	Line Material			316 S.S.				
	9	Line: Size, Schedule & I.D.			2 in		Sch. 40S I.D. 2.067		
	10	Equipment Number	Amb. Temp. Min. to Max.		20		to 95 °F		
	11	Tightness Requirements			ANSI IV (standard)				
	12	Allowable SPL dBA			85				
PROCESS	13	Fluid			Potable Water				
	14	Flow Operating			9962.44 lb/h				
	15	Flow	Max.	Min.	27348.2 lb/h		2001.2 lb/h		
	16	Inlet Pressure	Max.	Operating	136 psi-g		145 psi-g		
	17	DP Valve Sizing Operating			70 psi				
	18	DP Valve Sizing	Min.	Max.	73 psi		61 psi		
	19	Shut - Off DP	Max.		150 psi				
	20	Temperature	Max.	Operating	100 °F		88 °F		
	21	Spec. Gravity at Base	Operating				1		
	22	Molecular Mass							
	23	Viscosity at Base	Operating				0.78 cP		
	24	% Flashing							
	25	Vapor Pressure	Critical Pressure		0.656 psi-a		3206 psi-a		
	26	Design Temperature	Design Pressure		150 °F		150 psi-g		
BODY	27	Cv Calculated	Max.	Min.	7.0171		0.4679		
	28	Size	Trim Size	Cv	2 in		7.9		
	29	Form or Type			Standard				
	30	End Connections & Rating			Flanged, RF, Class 150				
	31	Body Material			ASME SA-216 WCC				
	32	Bonnet Type			N/A				
	33	Stem Material	Packing Material		17-4 PH				
	34	Flow Action to							
	35	Lubricator	Isolation Valve						
	36								
	TRIM	37	Guiding	Number Ports	Characteristic			Quick Opening	
		38	Plug Material	Seat Material		ASME SA-351 CF8M			
39		Ball Material	Gasket Material				PTFE		
40		Bushing Material							
ACTUATOR / PILOT	41								
	42	Model	Size		CCI		130		
	43	Type			Pressure Reducing Regulator				
	44	Pilot	Supply to Pilot		N/A		N/A		
	45	Self Connection	External Connection		Yes				
	46	Diaphragm Material			EPDM				
	47	Spring Range			66.72 : 101.53 psi.g				
	48	Set Point			75 psig				
	49								
OPTIONS	50	Filter Regulator							
	51	Line Strainer							
	52	Housing Vent							
	53	Internal Relief							
	54								
	55	Vendor	Price		IMI				
PURCHASE	56	Manufacturer			CARRARO				
	57	Model			UBAN-V/AF2/130				
	58	Detailed Model							
	59	Purchase Order Number							
	60	Serial Number							
	61								

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase		 Kiewit	
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
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No.	By	Chkd	Appr	Date	Revision	Dwg. No.: CP-139000-ICS-DAS-KIE-00001		Rev.: 1

Print Date: 14-Sep-2020 16:00:44

PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 55 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 148 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39052								
	2	Service			POT WAT TO GG CMPR AREA								
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001								
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3					
	5	Project Name		Project Number		Calcasieu Pass LNG		103930					
	6	Control Number		Work Breakdown Structure									
	7	Line Number			2"-139-000-PTW39059-1K0K3-1.0"HET								
	8	Line Material			316 S.S.								
	9	Line: Size, Schedule & I.D.			2		in		Sch. 40S		I.D. 2.067		
	10	Equipment Number		Amb. Temp. Min. to Max.			20		to 95		°F		
	11	Tightness Requirements			ANSI IV (standard)								
	12	Allowable SPL dBA			85								
PROCESS	13	Fluid			Potable Water								
	14	Flow Operating			9962.44		lb/h						
	15	Flow		Max. Min.		27348.2		lb/h		2001.2		lb/h	
	16	Inlet Pressure		Max. Operating		134		psi-g		143		psi-g	
	17	DP Valve Sizing Operating			68		psi						
	18	DP Valve Sizing		Min. Max.		72		psi		59		psi	
	19	Shut - Off DP		Max.		150		psi					
	20	Temperature		Max. Operating		100		°F		88		°F	
	21	Spec. Gravity at Base		Operating						1			
	22	Molecular Mass											
	23	Viscosity at Base		Operating						0.78		cP	
	24	% Flashing											
BODY	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a	
	26	Design Temperature		Design Pressure		150		°F		150		psi-g	
	27	Cv Calculated		Max. Min.		7.135				0.4712			
	28	Size		Trim Size		Cv		2		in		7.9	
	29	Form or Type			Standard								
	30	End Connections & Rating			Flanged, RF, Class 150								
	31	Body Material			ASME SA-216 WCC								
	32	Bonnet Type			N/A								
	33	Stem Material		Packing Material		17-4 PH							
	34	Flow Action to											
	35	Lubricator		Isolation Valve									
	36												
TRIM	37	Guiding		Number Ports		Characteristic				Quick Opening			
	38	Plug Material		Seat Material		ASME SA-351 CF8M							
	39	Ball Material		Gasket Material				PTFE					
	40	Bushing Material											
ACTUATOR / PILOT	41												
	42	Model		Size		CCI		130					
	43	Type			Pressure Reducing Regulator								
	44	Pilot		Supply to Pilot		N/A		N/A					
	45	Self Connection		External Connection		Yes							
	46	Diaphragm Material			EPDM								
	47	Spring Range			66.72 : 101.53 psi.g								
	48	Set Point			75 psig								
OPTIONS	49												
	50	Filter Regulator											
	51	Line Strainer											
	52	Housing Vent											
PURCHASE	53	Internal Relief											
	54												
	55	Vendor		Price		IMI							
	56	Manufacturer			CARRARO								
	57	Model			UBAN-V/AF2/130								
	58	Detailed Model											
	59	Purchase Order Number											
	60	Serial Number											
	61												

Notes: See notes

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					INSTRUMENT SPECIFICATION		VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase	Pressure Regulator		
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
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Print Date: 14-Sep-2020 16:00:45

PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 55 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 147 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39054								
	2	Service			POT WAT TO AMIN UNIT								
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001								
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3					
	5	Project Name		Project Number		Calcasieu Pass LNG		103930					
	6	Control Number		Work Breakdown Structure									
	7	Line Number			3"-139-000-PTW39011-1K0K3-1.0 "HET								
	8	Line Material			316 S.S.								
	9	Line: Size, Schedule & I.D.			3		in		Sch. 40S		I.D. 3.068		
	10	Equipment Number		Amb. Temp. Min. to Max.				20		to 95		°F	
	11	Tightness Requirements			ANSI IV (standard)								
	12	Allowable SPL dBA			85								
PROCESS	13	Fluid			Potable Water								
	14	Flow Operating			9962.44		lb/h						
	15	Flow		Max. Min.		27348.2		lb/h		2001.2		lb/h	
	16	Inlet Pressure		Max. Operating		93		psi-g		125		psi-g	
	17	DP Valve Sizing Operating			50		psi						
	18	DP Valve Sizing		Min. Max.		61		psi		18		psi	
	19	Shut - Off DP		Max.		150		psi					
	20	Temperature		Max. Operating		100		°F		88		°F	
	21	Spec. Gravity at Base		Operating						1			
	22	Molecular Mass											
	23	Viscosity at Base		Operating						0.78		cP	
	24	% Flashing											
	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a	
	26	Design Temperature		Design Pressure		150		°F		150		psi-g	
BODY	27	Cv Calculated		Max. Min.		12.9178				0.5122			
	28	Size		Trim Size		Cv		2		in		22.2	
	29	Form or Type			Standard								
	30	End Connections & Rating			Flanged, RF, Class 150								
	31	Body Material			ASME SA-216 WCC								
	32	Bonnet Type			N/A								
	33	Stem Material		Packing Material		17-4 PH							
	34	Flow Action to											
	35	Lubricator		Isolation Valve									
	36												
TRIM	37	Guiding		Number Ports		Characteristic						Quick Opening	
	38	Plug Material		Seat Material		ASME SA-479 316				ASME SA-479 316			
	39	Ball Material		Gasket Material						PTFE			
	40	Bushing Material											
41													
ACTUATOR / PILOT	42	Model		Size		CCI				140			
	43	Type			Pressure Reducing Regulator								
	44	Pilot		Supply to Pilot		N/A				N/A			
	45	Self Connection		External Connection		Yes							
	46	Diaphragm Material			EPDM								
	47	Spring Range			36.26 : 82.67 psi.g								
	48	Set Point			75 psig								
	49												
	50	Filter Regulator											
OPTIONS	51	Line Strainer											
	52	Housing Vent											
	53	Internal Relief											
	54												
PURCHASE	55	Vendor		Price		IMI							
	56	Manufacturer			CARRARO								
	57	Model			UBR-V/AF2/140								
	58	Detailed Model											
	59	Purchase Order Number											
	60	Serial Number											
	61												

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator Form: 209	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 1 of 2
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
No.	By	Chkd	Appr	Date	Revision	Dwg. No.: CP-139000-ICS-DAS-KIE-00001		Rev.: 1

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PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 55 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 136 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39055									
	2	Service			POT WAT TO LIQ UNIT S BLOCK									
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001									
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3						
	5	Project Name		Project Number		Calcasieu Pass LNG		103930						
	6	Control Number		Work Breakdown Structure										
	7	Line Number			3"-139-000-PTW39001-1K0K3-1.0 "HET									
	8	Line Material			316 S.S.									
	9	Line: Size, Schedule & I.D.			3		in		Sch. 40S		I.D. 3.068			
	10	Equipment Number		Amb. Temp. Min. to Max.				20		to 95		°F		
	11	Tightness Requirements			ANSI IV (standard)									
	12	Allowable SPL dBA			85									
PROCESS	13	Fluid			Potable Water									
	14	Flow Operating			9962.44		lb/h							
	15	Flow		Max. Min.		27348.2		lb/h		2001.2		lb/h		
	16	Inlet Pressure		Max. Operating		83		psi-g		120		psi-g		
	17	DP Valve Sizing Operating			45		psi							
	18	DP Valve Sizing		Min. Max.		57		psi		8		psi		
	19	Shut - Off DP		Max.		150		psi						
	20	Temperature		Max. Operating		100		°F		88		°F		
	21	Spec. Gravity at Base		Operating						1				
	22	Molecular Mass												
	23	Viscosity at Base		Operating						0.78		cP		
	24	% Flashing												
	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a		
	26	Design Temperature		Design Pressure		150		°F		150		psi-g		
BODY	27	Cv Calculated		Max. Min.		19.3769				0.5296				
	28	Size		Trim Size		Cv		2		in		35		
	29	Form or Type			Standard									
	30	End Connections & Rating			Flanged, RF, Class 150									
	31	Body Material			ASME SA-216 WCC									
	32	Bonnet Type			N/A									
	33	Stem Material		Packing Material		17-4 PH								
	34	Flow Action to												
	35	Lubricator		Isolation Valve										
	36													
	TRIM	37	Guiding		Number Ports		Characteristic				Quick Opening			
		38	Plug Material		Seat Material		ASME SA-479 316				ASME SA-479 316			
		39	Ball Material		Gasket Material						PTFE			
		40	Bushing Material											
41														
ACTUATOR / PILOT	42	Model		Size		CCI				140				
	43	Type			Back Pressure Regulator									
	44	Pilot		Supply to Pilot		N/A				N/A				
	45	Self Connection		External Connection		Yes								
	46	Diaphragm Material			EPDM									
	47	Spring Range			72.52 : 76.87 psi.g									
	48	Set Point			75 psig									
	49													
	50	Filter Regulator												
OPTIONS	51	Line Strainer												
	52	Housing Vent												
	53	Internal Relief												
	54													
PURCHASE	55	Vendor		Price		IMI								
	56	Manufacturer			CARRARO									
	57	Model			UBS-V/AF2/140									
	58	Detailed Model												
	59	Purchase Order Number												
	60	Serial Number												
	61													

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator Form: 209	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 1 of 2
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
No.	By	Chkd	Appr	Date	Revision	Dwg. No.: CP-139000-ICS-DAS-KIE-00001		Rev.: 1

Print Date: 14-Sep-2020 16:00:47

PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 55 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 132 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39056			
	2	Service			POT WAT TO BOG CMPR AREA			
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001			
	4	Location	Area Class		INLINE		Class I, Div 2, Group C & D, T3	
	5	Project Name	Project Number		Calcasieu Pass LNG		103930	
	6	Control Number	Work Breakdown Structure					
	7	Line Number			3"-139-000-PTW39061-1K0K3-1.0 "HET			
	8	Line Material			316 S.S.			
	9	Line: Size, Schedule & I.D.			3	in	Sch. 40S I.D. 3.068	
	10	Equipment Number	Amb. Temp. Min. to Max.		20		to 95 °F	
	11	Tightness Requirements			ANSI IV (standard)			
	12	Allowable SPL dBA			85			
PROCESS	13	Fluid			Potable Water			
	14	Flow Operating			9962.44 lb/h			
	15	Flow	Max.	Min.	37293	lb/h	2001.2 lb/h	
	16	Inlet Pressure	Max.	Operating	78	psi-g	118 psi-g	
	17	DP Valve Sizing Operating		43	psi			
	18	DP Valve Sizing	Min.	Max.	56	psi	3 psi	
	19	Shut - Off DP	Max.	150	psi			
	20	Temperature	Max.	Operating	100	°F	88 °F	
	21	Spec. Gravity at Base	Operating		1			
	22	Molecular Mass						
	23	Viscosity at Base		Operating	0.78 cP			
	24	% Flashing						
	25	Vapor Pressure	Critical Pressure		0.656	psi-a	3206 psi-a	
	26	Design Temperature	Design Pressure		150	°F	150 psi-g	
BODY	27	Cv Calculated	Max.	Min.	43.1515		0.5343	
	28	Size	Trim Size	Cv	2	in	52	
	29	Form or Type			Standard			
	30	End Connections & Rating			Flanged, RF, Class 150			
	31	Body Material			ASME SA-216 WCC			
	32	Bonnet Type			N/A			
	33	Stem Material	Packing Material		17-4 PH			
	34	Flow Action to						
	35	Lubricator	Isolation Valve					
	36							
	TRIM	37	Guiding	Number Ports	Characteristic	Quick Opening		
		38	Plug Material	Seat Material		ASME SA-479 316		ASME SA-479 316
		39	Ball Material	Gasket Material				PTFE
		40	Bushing Material					
41								
ACTUATOR / PILOT	42	Model	Size		CCI		182	
	43	Type			Back Pressure Regulator			
	44	Pilot	Supply to Pilot		N/A		N/A	
	45	Self Connection	External Connection		Yes			
	46	Diaphragm Material			EPDM			
	47	Spring Range			58.02 : 89.92 psi.g			
	48	Set Point			75 psig			
	49							
OPTIONS	50	Filter Regulator						
	51	Line Strainer						
	52	Housing Vent						
	53	Internal Relief						
	54							
	55	Vendor	Price		IMI			
PURCHASE	56	Manufacturer			CARRARO			
	57	Model			BPM84/AF2/182			
	58	Detailed Model						
	59	Purchase Order Number						
	60	Serial Number						
	61							

Notes: See notes

REV 209-12

REV 200-12						INSTRUMENT SPECIFICATION Pressure Regulator Form: 209		VENTURE GLOBAL LNG 		
1	MM	MA	EC	14-Sep-2020	Issued for Purchase					
0	MM	MA	EC	30-Jul-2020	Issued for Purchase					
B	MM	MA	EC	12-Jun-2020	Issued for Approval					
A	MM	MA	EC	11-Jun-2020	Issued for Review	Sheet 1		of 2		
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PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 75 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 131 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39057									
	2	Service			POT WAT TO LIQ UNIT N BLOCK									
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001									
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3						
	5	Project Name		Project Number		Calcasieu Pass LNG		103930						
	6	Control Number		Work Breakdown Structure										
	7	Line Number			3"-139-000-PTW39008-1K0K3-1.0 "HET									
	8	Line Material			316 S.S.									
	9	Line: Size, Schedule & I.D.			3		in		Sch. 40S		I.D. 3.068			
	10	Equipment Number		Amb. Temp. Min. to Max.				20		to 95		°F		
	11	Tightness Requirements			ANSI IV (standard)									
	12	Allowable SPL dBA			85									
PROCESS	13	Fluid			Potable Water									
	14	Flow Operating			34869.6		lb/h							
	15	Flow		Max. Min.		52210.2		lb/h		27016		lb/h		
	16	Inlet Pressure		Max. Operating		77		psi-g		117		psi-g		
	17	DP Valve Sizing Operating			42		psi							
	18	DP Valve Sizing		Min. Max.		55		psi		2		psi		
	19	Shut - Off DP		Max.		150		psi						
	20	Temperature		Max. Operating		100		°F		88		°F		
	21	Spec. Gravity at Base		Operating						1				
	22	Molecular Mass												
	23	Viscosity at Base		Operating						0.78		cP		
	24	% Flashing												
	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a		
	26	Design Temperature		Design Pressure		150		°F		150		psi-g		
BODY	27	Cv Calculated		Max. Min.		73.9831				7.2779				
	28	Size		Trim Size		Cv		3		in				
	29	Form or Type			Standard									
	30	End Connections & Rating			Flanged, RF, Class 150									
	31	Body Material			ASME SA-216 WCC									
	32	Bonnet Type			N/A									
	33	Stem Material		Packing Material		17-4 PH								
	34	Flow Action to												
	35	Lubricator		Isolation Valve										
	36													
	TRIM	37	Guiding		Number Ports		Characteristic				Quick Opening			
		38	Plug Material		Seat Material		ASME SA-479 316				ASME SA-479 316			
		39	Ball Material		Gasket Material						PTFE			
		40	Bushing Material											
ACTUATOR / PILOT	41													
	42	Model		Size		CCI				182				
	43	Type			Back Pressure Regulator									
	44	Pilot		Supply to Pilot		N/A				N/A				
	45	Self Connection		External Connection		Yes								
	46	Diaphragm Material			EPDM									
	47	Spring Range			50.75; 92.08 Psi.g									
	48	Set Point			75 psig									
	49													
OPTIONS	50	Filter Regulator												
	51	Line Strainer												
	52	Housing Vent												
	53	Internal Relief												
	54													
	PURCHASE	55	Vendor		Price		IMI							
56		Manufacturer			CARRARO									
57		Model			BPM82/AF2/182									
58		Detailed Model												
59		Purchase Order Number												
60		Serial Number												
61														

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator Form: 209	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 1 of 2
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
No.	By	Chkd	Appr	Date	Revision	Dwg. No.: CP-139000-ICS-DAS-KIE-00001		Rev.: 1

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PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 54 / 70 / 105 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 130 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
No.	By	Date	Revision	Code: 209	Dwg. No.: CP-139000-ICS-DAS-KIE-00001	Rev.: 1

GENERAL	1	Tag Number			139-000-PCV -39058								
	2	Service			POT WAT TO MERCURY REM UNIT								
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001								
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3					
	5	Project Name		Project Number		Calcasieu Pass LNG		103930					
	6	Control Number		Work Breakdown Structure									
	7	Line Number			3"-139-000-PTW39012-1K0K3-1.0 "HET								
	8	Line Material			316 S.S.								
	9	Line: Size, Schedule & I.D.			3		in		Sch. 40S		I.D. 3.068		
	10	Equipment Number		Amb. Temp. Min. to Max.				20		to 95		°F	
	11	Tightness Requirements			ANSI IV (standard)								
	12	Allowable SPL dBA			85								
PROCESS	13	Fluid			Potable Water								
	14	Flow Operating			9962.44		lb/h						
	15	Flow		Max. Min.		27348.2		lb/h		2001.2		lb/h	
	16	Inlet Pressure		Max. Operating		95		psi-g		127		psi-g	
	17	DP Valve Sizing Operating			52		psi						
	18	DP Valve Sizing		Min. Max.		61		psi		20		psi	
	19	Shut - Off DP		Max.		150		psi					
	20	Temperature		Max. Operating		100		°F		88		°F	
	21	Spec. Gravity at Base		Operating						1			
	22	Molecular Mass											
	23	Viscosity at Base		Operating						0.78		cP	
	24	% Flashing											
	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a	
	26	Design Temperature		Design Pressure		150		°F		150		psi-g	
BODY	27	Cv Calculated		Max. Min.		12.2549				0.5119			
	28	Size		Trim Size		Cv		2		in		22.2	
	29	Form or Type			Standard								
	30	End Connections & Rating			Flanged, RF, Class 150								
	31	Body Material			ASME SA-216 WCC								
	32	Bonnet Type			N/A								
	33	Stem Material		Packing Material		17-4 PH							
	34	Flow Action to											
	35	Lubricator		Isolation Valve									
	36												
	37	Guiding		Number Ports		Characteristic						Quick Opening	
	TRIM	38	Plug Material		Seat Material		ASME SA-479 316				ASME SA-479 316		
39		Ball Material		Gasket Material						PTFE			
40		Bushing Material											
41													
ACTUATOR / PILOT	42	Model		Size		CCI				140			
	43	Type			Pressure Reducing Regulator								
	44	Pilot		Supply to Pilot		N/A				N/A			
	45	Self Connection		External Connection		Yes							
	46	Diaphragm Material			EPDM								
	47	Spring Range			36.26 : 82.67 psi.g								
	48	Set Point			75 psig								
	49												
	50	Filter Regulator											
OPTIONS	51	Line Strainer											
	52	Housing Vent											
	53	Internal Relief											
	54												
PURCHASE	55	Vendor		Price		IMI							
	56	Manufacturer			CARRARO								
	57	Model			UBR-V/AF2/140								
	58	Detailed Model											
	59	Purchase Order Number											
	60	Serial Number											
	61												

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase			
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
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PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 55 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 136 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
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GENERAL	1	Tag Number			139-000-PCV -39059								
	2	Service			POT WAT HDR								
	3	P&ID Number			CP-139000-PRO-PID-KIE-00001-001								
	4	Location		Area Class		INLINE		Class I, Div 2, Group C & D, T3					
	5	Project Name		Project Number		Calcasieu Pass LNG							
	6	Control Number		Work Breakdown Structure									
	7	Line Number			3"-139-000-PTW39009-1K0K3-1.0 "HET								
	8	Line Material			316 S.S.								
	9	Line: Size, Schedule & I.D.			3		in		Sch. 40S		I.D. 3.068		
	10	Equipment Number		Amb. Temp. Min. to Max.				20		to 95		°F	
	11	Tightness Requirements			ANSI IV (standard)								
	12	Allowable SPL dBA			85								
PROCESS	13	Fluid			Potable Water								
	14	Flow Operating			9962.44		lb/h						
	15	Flow		Max. Min.		37293		lb/h		2001.2		lb/h	
	16	Inlet Pressure		Max. Operating		82		psi-g		127		psi-g	
	17	DP Valve Sizing Operating			52		psi						
	18	DP Valve Sizing		Min. Max.		62		psi		7		psi	
	19	Shut - Off DP		Max.		150		psi					
	20	Temperature		Max. Operating		100		°F		88		°F	
	21	Spec. Gravity at Base		Operating						1			
	22	Molecular Mass											
	23	Viscosity at Base		Operating						0.78		cP	
	24	% Flashing											
BODY	25	Vapor Pressure		Critical Pressure		0.656		psi-a		3206		psi-a	
	26	Design Temperature		Design Pressure		150		°F		150		psi-g	
	27	Cv Calculated		Max. Min.		28.2479				0.5078			
	28	Size		Trim Size		Cv		2		in		35	
	29	Form or Type			Standard								
	30	End Connections & Rating			Flanged, RF, Class 150								
	31	Body Material			ASME SA-216 WCC								
	32	Bonnet Type			N/A								
	33	Stem Material		Packing Material		17-4 PH							
	34	Flow Action to											
	35	Lubricator		Isolation Valve									
	36												
TRIM	37	Guiding		Number Ports		Characteristic				Quick Opening			
	38	Plug Material		Seat Material		ASME SA-479 316				ASME SA-479 316			
	39	Ball Material		Gasket Material						PTFE			
	40	Bushing Material											
41													
ACTUATOR / PILOT	42	Model		Size		CCI				140			
	43	Type			Back Pressure Regulator								
	44	Pilot		Supply to Pilot		N/A				N/A			
	45	Self Connection		External Connection		Yes							
	46	Diaphragm Material			EPDM								
	47	Spring Range			72.52 : 76.87 psi.g								
	48	Set Point			75 psig								
	49												
	50	Filter Regulator											
OPTIONS	51	Line Strainer											
	52	Housing Vent											
	53	Internal Relief											
	54												
PURCHASE	55	Vendor		Price		IMI							
	56	Manufacturer			CARRARO								
	57	Model			UBS-V/AF2/140								
	58	Detailed Model											
	59	Purchase Order Number											
	60	Serial Number											
	61												

Notes: See notes

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						INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	MA	EC	14-Sep-2020	Issued for Purchase			
0	MM	MA	EC	30-Jul-2020	Issued for Purchase			
B	MM	MA	EC	12-Jun-2020	Issued for Approval			
A	MM	MA	EC	11-Jun-2020	Issued for Review			
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PAU: Calcasieu Pass LNG/1/39 - Potable Water Systems

1. The valve shall comply with project specification "CP-100000-ICS-SPC-KIE-00002 - BOP Control Valves Specification" and CP-100000-PIP-SPC-KIE-00040 - "Specification of Valves for Non Cryogenic Services."
2. Testing of valve leakage shall be per ANSI FCI 70-2.
3. Paint coating shall meet C5M corrosion protection for marine environment.
4. Design Temperature (Min/Max): 20 / 150 deg. F.
5. Liquid Volumetric Flowrate (Min / Norm / Max): 4 / 20 / 75 gpm.
6. Liquid Upstream Density (Min / Norm / Max): 62.38 / 62.11 / 62 lb/ft3.
7. Temperature @ min. condition 60 deg. F.
8. Upstream Pressure @ min. condition 137 PSIG.
9. Liquid Specific Gravity (Min / Norm / Max): 1/1/0.99.
10. Liquid Viscosity (Min / Norm / Max): 1.12 / 0.78 / 0.68 cP.
11. Liquid Vapour Pressure (Min / Norm / Max): 0.256/ 0.656 / 0.951 PSIA.
12. VTA: Vendor to Advise.
13. Vendor to perform valve sizing and confirm valve size.

				INSTRUMENT SPECIFICATION Pressure Regulator	VENTURE GLOBAL LNG	
1	MM	14-Sep-2020	Issued for Purchase		 Kiewit	Sheet 2 of 2
0	MM	30-Jul-2020	Issued for Purchase			
B	MM	12-Jun-2020	Issued for Approval			
A	MM	11-Jun-2020	Issued for Review			
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